

Practice Test 2

Temporal Representation & Feedforward Transformations

Time: 30 minutes Total: 20 marks

Closed book. No calculator, formula sheet, or other devices. Show the requested reasoning. Exact symbolic answers and schematic sketches are sufficient; numerical evaluation of exponentials is unnecessary. Label all sketch axes. Reserve about four minutes to read and check.

Question 1. [2 marks]

A real, zero-mean random signal has duration 1 s, RMS 0.5, and a white spectrum bandlimited to 10 Hz. Use the notebook's Fourier-coefficient construction.

- (a) [1] Sketch its ensemble-average Fourier magnitude against angular frequency ω . Label both cutoff frequencies with units. Do not calculate the plateau height.
- (b) [1] The cutoff is doubled while RMS is kept fixed. Predict the change in the time trace and explain what does, and does not, change about its amplitude. Two sentences suffice.

Question 2. [2 marks]

Two normalized LIF neurons start with voltage zero. Their encoders are $e = +1$ and $e = -1$, with current $J = ex + 1$. Set the constant input to $x = 1/2$. Threshold is 1, reset is 0, $\tau_{RC} = 20$ ms, and $\tau_{\text{ref}} = 2$ ms. Negative voltages are clamped to zero.

Sketch both voltage traces against time. Show their initial evolution, threshold, and subsequent behaviour. Mark spikes and refractory intervals wherever they occur. Exact spike times are unnecessary.

Question 3. [7 marks]

(a) [2] Let $r(t) = a_+(t) - a_-(t)$ be the signed response of an opponent pair of spiking neurons; $X(\omega)$ and $R(\omega)$ are the input and response Fourier transforms. Write the ensemble-averaged optimal linear filter $H(\omega)$. Assume a bandlimited input and broadband response noise uncorrelated with it. On aligned frequency axes, sketch the qualitative spectral support of the input, response, and optimally decoded output; briefly justify the difference. Do not calculate heights.

(b) [3] A causal PSC kernel is

$$h_n(t) = \begin{cases} c_n^{-1} t^n e^{-t/\tau}, & t \geq 0, \\ 0, & t < 0, \end{cases}$$

where c_n makes its integral equal to one. Find c_0 and c_1 , showing the integrals or their evaluation. Sketch the two normalized kernels on common axes at the same τ , labelling their values at zero and the time of each peak.

(c) [2] With $n = 0$, explain two changes in the decoded time trace when τ increases. Also explain why a fixed PSC filter and fixed decoder typically track a 2 Hz bandlimited input more faithfully than a 10 Hz input of the same RMS. Both remain in the trained representation range. Distinguish signal distortion from residual spike noise. Three or four concise sentences suffice.

Question 4. [3 marks]

(a) [2] A population has n neurons representing a d -dimensional variable. The N rate-mode training samples are columns of $A \in \mathbb{R}^{n \times N}$ and $X \in \mathbb{R}^{d \times N}$. Write the noise-regularized identity decoder with penalty strength $N\sigma^2$, and state its dimensions. No inverse needs to be evaluated.

(b) [1] In a spiking simulation, bin k stores spike **counts** b_{ik} , rather than impulse heights. Samples h_k of a continuous unit-area PSC kernel have units s^{-1} . Write the filtered activity s_{ik} as a sum over spike

bins. State whether the sum needs an explicit Δt factor, explaining how it connects to the impulse-height convention before the rate decoder is reused. One equation and two short statements suffice.

Question 5.**[6 marks]**

(a) [2] Population A has n neurons. A function decoder $D^f \in \mathbb{R}^{q \times n}$, fitted over the whole input domain, estimates $f(\mathbf{x})$. Population B has m neurons representing $\mathbf{y} = f(\mathbf{x}) \in \mathbb{R}^q$, with unit encoders $\mathbf{e}_i$, gains α_i , and biases J_i^{bias} . Starting from the current encoding equation, derive the synaptic weight w_{ij} and matrix W^f , including its dimensions. Use rate activities. Three or four lines suffice.

(b) [1] For a scalar connection $y = 2x + 1$, population B was trained only on $[-1, 1]$. Its input is $x(t) = \frac{1}{2} \sin(2\pi t)$. Identify the desired output range and explain a resulting limitation in the decoded output. One range and one or two sentences suffice.

(c) [3] Draw labelled population-level networks for

$$z = 2y + \frac{1}{2}x, \quad p = xy.$$

External x and y initially have separate scalar populations. Label the dimension represented by every added population and the transformations on its connections. Under ordinary linear summation of decoded currents, explain why the architecture sufficient for the first computation is insufficient for the second. Two small diagrams and two sentences suffice; no weight matrices are required.

End of practice test — 20 marks.